Parallel and Distributed Computing — Calculus Edition

Field 80 of 80 — The LKVexa Compendium carried to the continuum

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Parallel and Distributed Computing — Calculus Image

Performance models are linear decompositions of a work vector — Amdahl and Gustafson differ only in what is held fixed, and the roofline is a minimum of two linear bounds. The algorithm section is where genuine linear algebra appears: scan is an associative-operator recursion, Cannon's algorithm is a communication-optimal matrix product, and the communication lower bound is a geometric argument about a three-dimensional iteration space. The consensus results at the end are combinatorial impossibility theorems, and they say so.

Each entry carries its originating equation from the Linear Algebra edition, its 8S image, its calculus image under $\mathsf{T}_{\mathrm{cal}}$, a six-step proof, and the original discussion. The schema (Q1)–(Q6), the master theorems Θ-A…Θ-J and the lemmas D1…D14 are established in `00_Foundations_Calculus_Transform.md`.

Performance Models

1. Speedup

Origin (LA edition).

$$S_p = \frac{T_1}{T_p}$$

Input (8S edition).

$$y_1(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_1)\Big] z(u), \qquad z_1(u) = \begin{bmatrix} c_1 \\ r_1 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_1)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_1\, \psi_1(c')\, p(c')\, dc', \qquad \partial_\tau \psi_1 = -\mathcal{L}_\star \psi_1, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so

no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_1 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_1)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(1)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_1$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio of two execution times.

Structure: Linear speedup $S_p = p$ is the ideal, and superlinear speedup indicates a cache effect — the aggregate cache grew with the processor count, so the comparison baseline changed. It is a ratio, so it does not compose additively.

Reduction: Speedup is a time ratio; superlinearity signals a changed baseline.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

2. Efficiency

Origin (LA edition).

$$E_p = \frac{S_p}{p}$$

Input (8S edition).

$$\left(g^{(5)}_{2j},\ \delta^{(8)}_{2j},\ g^{(3)}_{2j},\ g^{(\mathrm{J})}_{2j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{2j}, \qquad g^{(3)}_{2j} \leq g^{(5)}_{2j}$$

Calculus image.

$$d_g(z_2, z_j) = \inf_{\Gamma} \int_0^1 \|\dot\Gamma\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{2j} \leq d_g(z_2, z_j), \qquad g^{(3)}_{2j} \leq g^{(5)}_{2j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_2 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{2j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(2)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_2$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A normalised speedup.

Structure: Exactly speedup divided by resources, so it measures the fraction of ideal scaling achieved. Its product with p is the speedup, which makes iso-efficiency curves hyperbolas in the (p, S) plane.

Reduction: Efficiency normalises speedup by resource count.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

3. Amdahl's Law

Origin (LA edition).

$$S_p = \frac{1}{f + \frac{1-f}{p}}$$

Input (8S edition).

$$y_3(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_3)\Big] z(u), \qquad z_3(u) = \begin{bmatrix} c_3 \\ r_3 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_3)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_3\, \psi_3(c')\, p(c')\, dc', \qquad \partial_\tau \psi_3 = -\mathcal{L}_\star \psi_3, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_3 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_3)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(3)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_3$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A speedup bound from a fixed serial fraction.

Structure: Total work decomposes exactly into a serial and a parallel part — a two-component linear split of the work vector — and only the second divides. The bound $S_\infty = 1/f$ is the limit as the parallel term vanishes.

Reduction: The law is a two-component linear work decomposition.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

4. Gustafson–Barsis Law

Origin (LA edition).

$$S_p = p - f(p-1)$$

Input (8S edition).

$$\max_{n_4 \in \mathbb{Z}^5} \; \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_4\|_1 \leq M, \;\; \eta^{(4)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_4\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(4)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_4\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(4)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_4 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(4)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_4$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Speedup under a fixed execution time rather than fixed problem size.

Structure: Exactly affine in p, because the problem grows with the machine so the parallel work scales. Amdahl and Gustafson are the same decomposition with different quantities held constant — a change of the constraint, not of the model.

Reduction: Scaled speedup is affine in processor count.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

5. Karp–Flatt Metric

Origin (LA edition).

$$f_e = \frac{1/S_p - 1/p}{1 - 1/p}$$

Input (8S edition).

$$\left(g^{(5)}_{5j},\ \delta^{(8)}_{5j},\ g^{(3)}_{5j},\ g^{(\mathrm{J})}_{5j}\right) \;\longmapsto\; \mathcal{R}^{8S}_{5j}, \qquad g^{(3)}_{5j} \leq g^{(5)}_{5j}$$

Calculus image.

$$d_g(z_5, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{5j} \leq d_g(z_5, z_j), \qquad g^{(3)}_{5j} \leq g^{(5)}_{5j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_5 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point

is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{5j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(5)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_5$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An experimentally determined serial fraction.

Structure: Inverting Amdahl's law for f gives this expression, so an increasing f_e with p reveals overhead beyond the serial fraction. It is a diagnostic obtained by inverting a linear model.

Reduction: The metric inverts the Amdahl decomposition.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

6. Strong and Weak Scaling

Origin (LA edition).

fixed total size versus fixed size per processor

Input (8S edition).

$$y_6(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_6)\Big] z(u), \qquad z_6(u) = \begin{bmatrix} c_6 \\ r_6 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_6)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_6\, \psi_6(c')\, p(c')\, dc', \qquad \partial_\tau \psi_6 = -\mathcal{L}_\star \psi_6, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_6 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_6)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(6)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_6$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Two different limits of the same model.

Structure: Strong scaling holds the problem fixed and is bounded by Amdahl; weak scaling grows it and follows Gustafson. Which regime applies is a choice of how the work vector scales with p.

Reduction: The two regimes differ in how the work vector scales.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

7. Isoefficiency Function

Origin (LA edition).

$$W = KT_o(W, p)$$

Input (8S edition).

$$y_7(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_7)\Big] z(u), \qquad z_7(u) = \begin{bmatrix} c_7 \\ r_7 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_7)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_7\, \psi_7(c')\, p(c')\, dc', \qquad \partial_\tau \psi_7 = -\mathcal{L}_\star \psi_7, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_7 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_7)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(7)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_7$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An implicit relation defining how work must grow to hold efficiency.

Structure: Solving for W as a function of p gives the scalability signature — a linear isoefficiency means perfect scalability, higher powers mean poorer. It is an implicit function derived from the overhead's linear structure.

Reduction: Isoefficiency solves an implicit relation on the overhead.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

8. Cost and Cost Optimality

Origin (LA edition).

$$C = pT_p$$

Input (8S edition).

$$y_8(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_8)\Big] z(u), \qquad z_8(u) = \begin{bmatrix} c_8 \\ r_8 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_8)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_8\, \psi_8(c')\, p(c')\, dc', \qquad \partial_\tau \psi_8 = -\mathcal{L}_\star \psi_8, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_8 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_8)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(8)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_8$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A processor–time product.

Structure: Cost optimal means $pT_p = \Theta(T_1)$ — no asymptotic work is wasted, so efficiency stays bounded away from zero. It is a product of two quantities, hence additive in logs.

Reduction: Cost is a product; optimality bounds it against the serial work.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

9. Overhead Function

Origin (LA edition).

$$T_o = pT_p - T_1$$

Input (8S edition).

$$y_9(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_9)\Big] z(u), \qquad z_9(u) = \begin{bmatrix} c_9 \\ r_9 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_9)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_9\, \psi_9(c')\, p(c')\, dc', \qquad \partial_\tau \psi_9 = -\mathcal{L}_\star \psi_9, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_9 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_9)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(9)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_9$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An exact linear difference measuring wasted work.

Structure: Includes communication, idling, and excess computation, all of which add — so the overhead is a sum of independent contributions. Modelling each term separately is what makes performance prediction tractable.

Reduction: Overhead is a linear difference decomposing into additive sources.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

10. Roofline Model

Origin (LA edition).

$$P = \min(P_{\text{peak}},\ I \times B)$$

Input (8S edition).

$$W_{10j} = A_{10j} \exp\left[-\frac{(c_{10} - c_j)^\top G_5 (c_{10} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(10, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 10} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{10} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|_{G_5}^2 / 2\sigma_c^2 = \|x\|_{G_5}^2 / 4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi / 4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(10)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{10}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A minimum of two linear bounds.

Structure: Piecewise linear in the arithmetic intensity, with a single kink at the machine balance point P_{peak}/B. Whether a kernel is compute- or memory-bound is decided by which side of that kink it falls on.

Reduction: The roofline is a minimum of two linear functions.

Status: Exact linear (piecewise).

8S status: Exact linear (piecewise); preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear (piecewise); carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

11. Arithmetic Intensity

Origin (LA edition).

$$I = \frac{\text{FLOPs}}{\text{bytes moved}}$$

Input (8S edition).

$$W_{11j} = A_{11j} \exp\left[-\frac{(c_{11} - c_j)^\top G_5 (c_{11} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(11, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 11} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{11} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|_{G_5}^2/2\sigma_c^2 = \|x\|_{G_5}^2/4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi/4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(11)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{11}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio determining the roofline position.

Structure: Matrix multiplication has $I = O(n)$ and can be tiled to approach peak, while a vector operation has $I = O(1)$ and is bandwidth-bound. That difference is the reason dense linear algebra dominates benchmark results.

Reduction: Intensity is a work-to-traffic ratio positioning the kernel on the roofline.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

12. Little's Law for Concurrency

Origin (LA edition).

$$\text{concurrency} = \text{throughput} \times \text{latency}$$

Input (8S edition).

$$y_{12}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{12})\Big] z(u), \qquad z_{12}(u) = \begin{bmatrix} c_{12} \\ r_{12} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{12})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{12}\, \psi_{12}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{12} = -\mathcal{L}_\star \psi_{12}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{12} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{12})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(12)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{12}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A stock–flow identity.

Structure: Exactly linear and distribution-free, requiring only stability — so hiding latency requires proportionally more concurrency. It is the same identity as in queueing and in epidemiology's prevalence relation.

Reduction: The law is a linear stock–flow identity.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Operations Research.

Communication

13. Communication Time Model

Origin (LA edition).

$$T = \alpha + \beta m$$

Input (8S edition).

$$y_{13}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{13})\Big] z(u), \qquad z_{13}(u) = \begin{bmatrix} c_{13} \\ r_{13} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{13})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{13}\, \psi_{13}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{13} = -\mathcal{L}_\star \psi_{13}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{13} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{13})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(13)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{13}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An affine cost in message size.

Structure: Latency α and inverse bandwidth β are the intercept and slope, so message aggregation trades many intercepts for one. The crossover size α/β tells you when to combine messages.

Reduction: Communication cost is affine in message size.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

14. LogP Model

Origin (LA edition).

$$L,\ o,\ g,\ P$$

Input (8S edition).

$$y_{14}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{14})\Big]z(u), \qquad z_{14}(u) = \begin{bmatrix} c_{14} \\ r_{14}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{14})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{14}\, \psi_{14}(c')\, p(c')\, dc', \qquad \partial_\tau\psi_{14} = -\mathcal{L}_\star\psi_{14}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{14} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{14})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(14)}_{\text{ind}})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{14}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A four-parameter refinement of entry 13.

Structure: Separates latency, overhead, gap, and processor count, so the total cost is a linear combination of these with problem-dependent coefficients. It captures pipelining limits that the two-parameter model misses.

Reduction: LogP is a four-term linear cost model.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

15. BSP Superstep Cost

Origin (LA edition).

$$T = w + hg + l$$

Input (8S edition).

$$y_{15}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{15})\Big] z(u), \qquad z_{15}(u) = \begin{bmatrix} c_{15} \\ r_{15} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{15})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{15}\, \psi_{15}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{15} = -\mathcal{L}_\star \psi_{15}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{15} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{15})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(15)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{15}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A sum of computation, communication, and synchronisation.

Structure: Exactly additive across the three phases, and the total program cost is the sum over supersteps — a linear accumulation. The bulk-synchronous structure is what makes cost composition additive at all.

Reduction: BSP cost is an additive three-term model, summed over supersteps.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

16. Bisection Bandwidth

Origin (LA edition).

$$\text{minimum bandwidth across any halving cut}$$

Input (8S edition).

$$\max_{n_{16}\in\mathbb{Z}^5} \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{16}\|_1 \le M, \;\; \eta_{\text{ind}}^{(16)} \ge \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{16}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(16)})\Big] = 0, \qquad \lambda,\nu \ge 0, \qquad \lambda(\|n_{16}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(16)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{16} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \ge 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(16)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{16}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A minimum cut of the network graph.

Structure: Exactly a graph min-cut, and by max-flow min-cut duality it bounds the achievable all-to-all throughput. It is the single most important network metric for communication-heavy workloads.

Reduction: Bisection bandwidth is a min-cut; duality bounds throughput.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

17. Network Diameter and Degree

Origin (LA edition).

$$\text{hops} \le d, \qquad \text{ports per node}$$

Input (8S edition).

$$\Big(g^{(5)}_{17j},\ \delta^{(8)}_{17j},\ g^{(3)}_{17j},\ g^{(\mathrm{J})}_{17j}\Big)\ \longmapsto\ \mathcal{R}^{8S}_{17j}, \qquad g^{(3)}_{17j} \le g^{(5)}_{17j}$$

Calculus image.

$$d_g(z_{17}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot\Gamma\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{17j} \le d_g(z_{17}, z_j), \qquad g^{(3)}_{17j} \le g^{(5)}_{17j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{17} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{17j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(17)}_{\mathrm{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{17}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Two properties of the topology graph.

Structure: The Moore bound relates them — a degree-k diameter-d graph has at most $O(k^d)$ nodes, a counting argument. Topologies trade degree against diameter, and the diameter is the distance matrix's maximum entry.

Reduction: Diameter is a distance matrix maximum; the Moore bound is a count.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

18. Fat Tree Bandwidth

Origin (LA edition).

constant bisection bandwidth per level

Input (8S edition).

$$y_{18}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{18})\Big] z(u), \qquad z_{18}(u) = \begin{bmatrix} c_{18} \\ r_{18} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{18})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{18}\, \psi_{18}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{18} = -\mathcal{L}_\star \psi_{18}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{18} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{18})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(18)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{18}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A topology designed to keep the min-cut constant.

Structure: Increasing link capacity toward the root keeps the bisection bandwidth proportional to the node count — so the cut value is designed rather than emergent. Tapered fat trees trade cost for reduced bisection.

Reduction: The topology engineers the min-cut value.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

19. Dragonfly Topology

Origin (LA edition).

groups connected all-to-all

Input (8S edition).

$$y_{19}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{19})\Big]z(u), \qquad z_{19}(u) = \begin{bmatrix} c_{19} \\ r_{19}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{19})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{19}\, \psi_{19}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{19} = -\mathcal{L}_\star \psi_{19}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{19} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{19})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(19)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{19}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A hierarchical graph with small diameter.

Structure: Two levels of all-to-all give diameter three at modest degree, and adaptive routing spreads load across paths — a flow assignment problem on the graph. It exploits high-radix switches to approach the Moore bound.

Reduction: The topology is a hierarchical graph; routing is a flow assignment.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

20. Collective Communication Cost

Origin (LA edition).

$$T_{\text{allreduce}} \approx 2(p-1)\left(\alpha + \frac{m\beta}{p}\right)$$

Input (8S edition).

$$\max_{n_{20}\in\mathbb{Z}^5} \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{20}\|_1 \leq M,\ \ \eta^{(20)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{20}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(20)}_{\text{ind}})\Big] = 0, \qquad \lambda,\nu \geq 0, \qquad \lambda(\|n_{20}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(20)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{20} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(20)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{20}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A cost model for a structured communication pattern.

Structure: The ring algorithm's cost is exactly linear in p for latency and independent of p for bandwidth, so it is bandwidth-optimal. Recursive doubling gives $\log p$ latency instead, and the choice depends on where the message size sits relative to α/β.

Reduction: Collective cost is a linear model; algorithm choice optimises the trade.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

21. Broadcast and Scatter Costs

Origin (LA edition).

$$T_{\text{bcast}} \approx \log p(\alpha + m\beta)$$

Input (8S edition).

$$y_{21}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{21})\Big] z(u), \qquad z_{21}(u) = \begin{bmatrix} c_{21} \\ r_{21} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{21})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{21}\, \psi_{21}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{21} = -\mathcal{L}_\star \psi_{21}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so

no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{21} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{21})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(21)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{21}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A tree-structured communication cost.

Structure: Binomial tree broadcast costs $\log p$ steps, while scatter–allgather costs $2 \log p$ latency but only $m\beta$ bandwidth — the same two-regime structure as entry 20. Message size selects the algorithm.

Reduction: Broadcast cost is a tree depth times a linear message cost.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

22. Bandwidth–Latency Product

Origin (LA edition).

$$\text{bytes in flight} = B \times L$$

Input (8S edition).

$$y_{22}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{22})\Big] z(u), \qquad z_{22}(u) = \begin{bmatrix} c_{22} \\ r_{22} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{22})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{22}\, \psi_{22}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{22} = -\mathcal{L}_\star \psi_{22}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{22} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{22})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(22)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{22}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A stock–flow product — Little's law again.

Structure: Exactly the number of bytes that must be outstanding to saturate the link, so deep pipelining is required on high-bandwidth long-latency networks. It sets the minimum buffer size.

Reduction: The product is a stock–flow identity.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

23. Communication–Computation Overlap

Origin (LA edition).

$$T = \max(T_{\text{comp}}, T_{\text{comm}}) \text{ ideally}$$

Input (8S edition).

$$y_{23}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{23})\Big] z(u), \qquad z_{23}(u) = \begin{bmatrix} c_{23} \\ r_{23} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{23})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{23}\, \psi_{23}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{23} = -\mathcal{L}_\star \psi_{23}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{23} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{23})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(23)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{23}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A maximum rather than a sum.

Structure: Perfect overlap replaces addition with maximum — a change from ℓ_1 to ℓ_∞ composition, which is exactly the difference between serial and parallel resource use. Real overlap is partial, giving something between.

Reduction: Overlap replaces a sum by a maximum.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

24. Communication Lower Bound

Origin (LA edition).

$$Q = \Omega\left(\frac{n^3}{\sqrt{M}}\right)$$

Input (8S edition).

$$P_5^{(24)} = C_5 C_5^{+} = P_4 + \frac{w_{24} w_{24}^\top}{\|w_{24}\|_2^2}, \qquad w_{24} = (I - C_4 C_4^{+}) e_{24}, \qquad \eta_{\text{ind}}^{(24)} = \frac{\|w_{24}\|_2^2}{\|e_{24}\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_{24} - P_{\mathcal{V}} e_{24},\ v \rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta_{\text{ind}}^{(24)} = \frac{\det \mathrm{Gr}(C_4, e_{24})}{\det \mathrm{Gr}(C_4)(\|e_{24}\|^2 + \varepsilon)}, \qquad \nabla_{e_{24}} \|w_{24}\|^2 = 2\, w_{24}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_{24} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is

automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_{24}\|^2/(\|e_{24}\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(24)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{24}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A geometric bound on data movement for matrix multiplication.

Structure: The Loomis–Whitney inequality bounds the volume of a three-dimensional iteration space by the product of its three projections' areas, so a working set of size M can cover only $M^{3/2}$ operations. That geometric inequality gives the tile size $\sqrt{M/3}$ and hence the bound — one of the most elegant results in the field, and it is pure geometry.

Reduction: The bound is Loomis–Whitney on the iteration space's projections.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

Memory Hierarchy

25. Average Memory Access Time

Origin (LA edition).

$$\text{AMAT} = t_{\text{hit}} + m\, t_{\text{penalty}}$$

Input (8S edition).

$$y_{25}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{25})\Big] z(u), \qquad z_{25}(u) = \begin{bmatrix} c_{25} \\ r_{25} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{25})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{25}\, \psi_{25}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{25} = -\mathcal{L}_\star \psi_{25}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{25} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs

(R1) and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{25})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(25)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{25}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An affine combination nested across levels.

Structure: Each level's penalty is the next level's AMAT, so the total is a nested chain of affine maps — a continued composition. Miss rates multiply down the hierarchy.

Reduction: AMAT is a nested affine composition.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

26. Cache Miss Classification

Origin (LA edition).

compulsory, capacity, conflict, coherence

Input (8S edition).

$$y_{26}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{26})\Big]z(u), \qquad z_{26}(u) = \begin{bmatrix} c_{26} \\ r_{26}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{26})(c) = \int_{\mathcal{C}} \gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{26}\,\psi_{26}(c')\,p(c')\,dc', \qquad \partial_\tau\psi_{26} = -\mathcal{L}_\star\psi_{26}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{26}\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{26})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(26)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{26}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An additive attribution of misses.

Structure: Measured by differencing against idealised caches, so the categories are defined by subtraction — approximately additive but not orthogonal. Coherence misses are the parallel-specific addition.

Reduction: The categories are difference-based attributions.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

27. False Sharing Cost

Origin (LA edition).

cache line ping-pong between cores

Input (8S edition).

$$y_{27}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{27})\Big] z(u), \qquad z_{27}(u) = \begin{bmatrix} c_{27} \\ r_{27} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{27})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{27}\, \psi_{27}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{27} = -\mathcal{L}_\star \psi_{27}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{27} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind.

Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{27})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(27)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{27}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Coherence traffic from unrelated data sharing a line.

Structure: The unit of coherence is the line, not the word, so the sharing pattern is determined by the address-to-line map — a coarsening. Padding to line boundaries is a change of the data layout that decouples the mapping.

Reduction: False sharing arises from a coarsening of the address map.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

28. Cache Coherence Protocol States

Origin (LA edition).

MESI: Modified, Exclusive, Shared, Invalid

Input (8S edition).

$$\text{Admit}_{28}(8S) \iff \eta_{\text{ind}}^{(28)} \geq \varepsilon_{\text{ind}} \;\wedge\; \Delta_{8S} > \varepsilon_{\text{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \;\text{ or }\; \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{28}(x, \lambda) = 0, \qquad \partial_x \mathcal{G}_{28} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x \mathcal{G}_{28} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{28} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x, \lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with

the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(28)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{28}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A finite state machine per cache line.

Structure: Exactly a labelled transition system, and its correctness is a reachability property — the invariant is that at most one cache holds a line in Modified state. Verification is model checking on the state graph.

Reduction: Coherence is a finite state machine with a reachability invariant.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear; carried to the continuum by Lemma D7, and the input was algebraic, so the differential content is entirely supplied by the transform.

29. Directory Protocol Scalability

Origin (LA edition).

$$O(p) \text{ bits per line for a full bit vector}$$

Input (8S edition).

$$E^{8S}_{29}(z) = z^{\top}\mathcal{G}_8 z\Big|_{\text{node } 29}, \qquad \big\|z_{29}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta^{(8)}_{29j} = \sqrt{(c_{29}-c_j)^{\top}G_5(c_{29}-c_j) + (r_{29}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{29}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{29} = 0 \iff \mathcal{L}_{\star}\phi = 0, \qquad ds^2_{29} = dc^{\top}(G_5 + \nabla r\, \nabla r^{\top})dc + r^2_{29}\, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{29} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_{\star}\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^{\top}\nabla r \nabla r^{\top} dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^{\top}\mathcal{L}_{8S}x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(29)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{29}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A sharer set representation.

Structure: A full bit vector costs p bits per line, so storage is quadratic in the machine size — limited sharer lists trade exactness for space. It is a sparse representation trade-off on a sharing matrix.

Reduction: The directory is a sharing matrix; limited lists are a sparse approximation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

30. NUMA Access Ratio

Origin (LA edition).

$$\frac{t_{\text{remote}}}{t_{\text{local}}}$$

Input (8S edition).

$$\max_{n_{30}\in\mathbb{Z}^5} \ \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{30}\|_1 \le M, \ \ \eta_{\text{ind}}^{(30)} \ge \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{30}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(30)})\Big] = 0, \qquad \lambda,\nu \ge 0, \qquad \lambda(\|n_{30}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(30)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{30} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \ge 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(30)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{30}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio of two access latencies.

Structure: The full latency structure is a matrix over node pairs, and placement optimisation minimises the traffic-weighted sum of its entries — a quadratic assignment problem. First-touch policy is a greedy heuristic for it.

Reduction: NUMA placement minimises a traffic-weighted latency matrix contraction.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

31. Memory Bandwidth per Core

Origin (LA edition).

$$\frac{B_{\text{total}}}{n_{\text{cores}}}$$

Input (8S edition).

$$W_{31j} = A_{31j} \exp\left[-\frac{(c_{31} - c_j)^\top G_5 (c_{31} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(31, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 31} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{31} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|_{G_5}^2 / 2\sigma_c^2 = \|x\|_{G_5}^2 / 4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi / 4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(31)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{31}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A per-core share of a shared resource.

Structure: Bandwidth grows more slowly than core count, so the per-core share falls — this is the memory wall, and it moves kernels leftward on the roofline. It is a simple division with severe consequences.

Reduction: Per-core bandwidth is a shared resource division.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

32. Cache-Oblivious Complexity

Origin (LA edition).

$$O\left(\frac{n}{B}\log_{M/B}\frac{n}{B}\right)$$

Input (8S edition).

$$E_{32}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node }32}, \qquad \|z_{32}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta_{32j}^{(8)} = \sqrt{(c_{32}-c_j)^\top G_5 (c_{32}-c_j) + (r_{32}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{32}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{32} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{32}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{32}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{32} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(32)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{32}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An optimal transfer count achieved without tuning.

Structure: Recursive divide-and-conquer is self-similar at every scale, so it automatically matches whatever cache sizes exist — the recursion's scale invariance is what makes it oblivious. It is the same fractal argument as in cache-oblivious matrix multiplication.

Reduction: Optimality follows from the recursion's scale invariance.

Status: Structural analogue.

8S status: Structural analogue; the 8S image is likewise structural, and admission is conditional (Lemma T3).

Calculus status: Structural analogue; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

Parallel Algorithms

33. PRAM Work–Depth

Origin (LA edition).

$$T_p \le \frac{W}{p} + D$$

Input (8S edition).

$$y_{33}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{33})\Big] z(u), \qquad z_{33}(u) = \begin{bmatrix} c_{33} \\ r_{33} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{33})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{33}\, \psi_{33}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{33} = -\mathcal{L}_\star \psi_{33}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{33} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{33})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(33)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{33}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Brent's theorem — a linear bound in two quantities.

Structure: Exactly affine in the work and the depth, so the two quantities fully characterise a computation's parallel potential. It is the fundamental theorem of parallel algorithm analysis.

Reduction: Brent's bound is affine in work and depth.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

34. Parallelism and Span

Origin (LA edition).

$$\bar{P} = \frac{W}{D}$$

Input (8S edition).

$$\mathrm{Admit}_{34}(8S) \iff \eta_{\mathrm{ind}}^{(34)} \geq \varepsilon_{\mathrm{ind}} \ \wedge \ \Delta_{8S} > \varepsilon_{\mathrm{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \ \text{ or } \ \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{34}(x,\lambda) = 0, \qquad \partial_x \mathcal{G}_{34} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x \mathcal{G}_{34} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{34} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x,\lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(34)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{34}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio bounding useful processor count.

Structure: Adding processors beyond W/D gives no further speedup, since the critical path dominates — so the ratio is the maximum exploitable parallelism. The span is the longest path in the computation DAG.

Reduction: Parallelism is a work-to-critical-path ratio.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear; carried to the continuum by Lemma D7, and the input was algebraic, so the differential content is entirely supplied by the transform.

35. Parallel Prefix (Scan)

Origin (LA edition).

$$y_i = x_1 \oplus x_2 \oplus \cdots \oplus x_i$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta_{\text{ind}}^{(35)} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_{35}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{35} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(35)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{35}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A cumulative application of an associative operator.

Structure: Associativity permits a tree reduction and a downward sweep, giving $O(\log n)$ depth with $O(n)$ work — Blelloch's algorithm. Scan is exactly a triangular linear operator when $\oplus$ is addition, and the parallel algorithm is a factorisation of that triangular matrix into $\log n$ sparse factors.

Reduction: Scan is a triangular operator factorised into $\log n$ sparse steps.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input was algebraic, so the differential content is entirely supplied by the transform.

36. Parallel Reduction

Origin (LA edition).

$$O(\log p) \text{ depth}$$

Input (8S edition).

$$y_{36}(\tau,u)=\Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{36})\Big]z(u),\qquad z_{36}(u)=\begin{bmatrix}c_{36}\\ r_{36}u\end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{36})(c)=\int_{\mathcal{C}}\gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{36}\,\psi_{36}(c')\,p(c')\,dc',\qquad \partial_\tau\psi_{36}=-\mathcal{L}_\star\psi_{36},\qquad \mathcal{L}_\star=-\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{36}\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du=\sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{36})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S}-\nu(\varepsilon_{\mathrm{ind}}-\eta_{\mathrm{ind}}^{(36)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{36}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A tree contraction of an associative operator.

Structure: Exactly a binary tree of pairwise combinations, so depth is logarithmic and work is linear. Floating-point addition is not truly associative, which is why reduction results depend on the tree shape — a genuine reproducibility issue caused by an algebraic assumption failing.

Reduction: Reduction is a tree contraction; non-associativity of floating point breaks determinism.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

37. Parallel Sorting Lower Bound

Origin (LA edition).

$$\Omega\left(\frac{n\log n}{p}+\log n\right)$$

Input (8S edition).

$$\max_{n_{37}\in\mathbb{Z}^5} \Delta_{8S} = \mathrm{Score}(\mathcal{M}_8) - \mathrm{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{37}\|_1 \le M, \;\; \eta_{\mathrm{ind}}^{(37)} \ge \varepsilon_{\mathrm{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{37}\|_1 - M) - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(37)})\Big] = 0, \qquad \lambda,\nu \ge 0, \qquad \lambda(\|n_{37}\|_1 - M) = \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(37)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{37} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \ge 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(37)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{37}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A bound combining work and depth terms.

Structure: The first term is the work bound divided by processors and the second is the unavoidable depth — exactly Brent's structure. It is the sum of two independent constraints.

Reduction: The bound is a sum of a work and a depth term.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

38. Bitonic Sort Complexity

Origin (LA edition).

$$O(\log^2 n) \text{ depth}$$

Input (8S edition).

$$y_{38}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{38})\Big] z(u), \qquad z_{38}(u) = \begin{bmatrix} c_{38} \\ r_{38} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_{\sigma}\psi_{38})(c) = \int_{\mathcal{C}} \gamma(c)\, W_{\sigma}(c,c')\, \gamma(c')\, \mathsf{A}_{38}\, \psi_{38}(c')\, p(c')\, dc', \qquad \partial_{\tau}\psi_{38} = -\mathcal{L}_{\star}\psi_{38}, \qquad \mathcal{L}_{\star} = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{38} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_{\sigma}$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_{\star}$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{38})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi\to[\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(38)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{38}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A sorting network with a fixed comparison pattern.

Structure: The comparison pattern is data-independent, so it is a fixed sequence of permutation-and-compare stages — a product of structured operators. That obliviousness makes it ideal for GPUs and for hardware, despite the extra $\log n$ factor.

Reduction: The network is a fixed product of compare-exchange operators.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

39. Matrix Multiplication (Cannon's Algorithm)

Origin (LA edition).

skew, then shift and multiply

Input (8S edition).

$$\mathcal{Y}_{39}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{39})\Big]z(u), \qquad z_{39}(u) = \begin{bmatrix} c_{39} \\ r_{39}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_{\sigma}\psi_{39})(c) = \int_{\mathcal{C}} \gamma(c)\, W_{\sigma}(c,c')\, \gamma(c')\, \mathsf{A}_{39}\, \psi_{39}(c')\, p(c')\, dc', \qquad \partial_{\tau}\psi_{39} = -\mathcal{L}_{\star}\psi_{39}, \qquad \mathcal{L}_{\star} = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{39} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_{\sigma}$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_{\star}$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{39})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(39)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{39}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A systolic algorithm on a $\sqrt{p}\times\sqrt{p}$ processor grid.

Structure: Initial skewing is a permutation of the block matrices, then each step is a local multiply and a cyclic shift — so the whole algorithm is a sequence of permutations and block products. It achieves the communication lower bound of entry 24, using $O(n^2/\sqrt{p})$ words per processor.

Reduction: Cannon's algorithm is permutations plus block products; it attains the lower bound.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

40. Parallel FFT

Origin (LA edition).

$$O\left(\frac{n\log n}{p} + \text{communication}\right)$$

Input (8S edition).

$$\widehat{W}^{(40)}(k) = \sum_{m} W(m)\, e^{-2\pi i\langle k,m\rangle/L}, \qquad F_5 = F^{\otimes 5}, \qquad F_5 W F_5^{*} = \mathrm{diag}(\widehat{W}(k))$$

Calculus image.

$$\widehat{W_\sigma * \psi}_{40}(\xi) = \widehat{W_\sigma}(\xi)\,\widehat{\psi_{40}}(\xi), \qquad \widehat{W_\sigma}(\xi) = e^{-\sigma_c^2 \xi^\top G_5^{-1} \xi/2}, \qquad \widehat{\partial_a \psi_{40}}(\xi) = i\,\xi_a\,\widehat{\psi_{40}}(\xi)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T12**, so the calculus image is of class **D12** (Fourier transform and pseudodifferential symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fourier transform and pseudodifferential symbol of Lemma D12, instantiated at the point $c_{40} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D12 needs $W_\sigma \in L^1$ for the convolution theorem, which the Gaussian satisfies, and ψ in a space where the transform is defined; $L^1 \cap L^2$ suffices and is standing. Translation invariance of the mask is required for W_σ to act as a convolution at all.

(Q4) Derivation. Apply the convolution theorem, then evaluate the transform of the Gaussian by completing the square in the exponent: $\int e^{-x^\top A x} e^{-i\xi\cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi/4}$ with $A = G_5/2\sigma_c^2$. The differentiation rule follows by integrating by parts in the transform integral, the boundary term vanishing by decay. Together these exhibit the coupling as a pseudodifferential operator with Gaussian symbol.

(Q5) Retraction through the tower. On the lattice torus the Fourier transform becomes the five-fold DFT $F_5 = F^{\otimes 5}$ and the symbol becomes the eigenvalue $\widehat{W}(k)$ of the 8S circulant. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(40)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{40}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A distributed butterfly network computation.

Structure: The FFT is a factorisation of the DFT matrix into $\log n$ sparse factors, and the parallel version distributes those factors — but the transpose steps require an all-to-all, which dominates the cost. Communication, not arithmetic, is the bottleneck.

Reduction: The FFT factorises the DFT matrix; distribution makes the transpose dominant.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T12.

Calculus status: Exact linear; carried to the continuum by Lemma D12, and the input was algebraic, so the differential content is entirely supplied by the transform.

41. Domain Decomposition Surface-to-Volume

Origin (LA edition).

$$\frac{\text{communication}}{\text{computation}} \propto \frac{1}{n^{1/d}}$$

Input (8S edition).

$$\left(g_{41j}^{(5)},\ \delta_{41j}^{(8)},\ g_{41j}^{(3)},\ g_{41j}^{(\mathrm{J})}\right) \;\longmapsto\; \mathcal{R}_{41j}^{8S}, \qquad g_{41j}^{(3)} \leq g_{41j}^{(5)}$$

Calculus image.

$$d_g(z_{41}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta_{41j}^{(8)} \leq d_g(z_{41}, z_j), \qquad g_{41j}^{(3)} \leq g_{41j}^{(5)}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{41} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{41j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(41)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{41}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A geometric scaling of the halo exchange.

Structure: Computation scales with the subdomain volume and communication with its surface, so the ratio falls as subdomains grow — which is why weak scaling works better than strong. It is a pure geometry argument.

Reduction: The ratio is a surface-to-volume geometric scaling.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

42. Parallel Monte Carlo Efficiency

Origin (LA edition).

near-perfect scaling with independent streams

Input (8S edition).

$$E^{8S}_{42}(z) = z^{\top}\mathcal{G}_8 z\Big|_{\text{node } 42}, \qquad \big\|z_{42}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta^{(8)}_{42j} = \sqrt{(c_{42}-c_j)^{\top}G_5(c_{42}-c_j) + (r_{42}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{42}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{42} = 0 \iff \mathcal{L}_\star\phi = 0, \qquad ds^2_{42} = dc^{\top}(G_5 + \nabla r\, \nabla r^{\top})dc + r^2_{42}\, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{42} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(42)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{42}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An embarrassingly parallel computation with a variance bound.

Structure: Independent samples mean variances add, so the error falls as $1/\sqrt{N}$ regardless of how the samples are distributed — the estimator is a linear functional and the parallelism is free. Stream independence is the only correctness requirement.

Reduction: Independent sampling makes the estimator a trivially decomposable sum.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

43. Speculative Execution Gain

Origin (LA edition).

$$E[\text{gain}] = p\, g - (1-p)c$$

Input (8S edition).

$$E^{8S}_{43}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 43}, \qquad \big\|z_{43}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta^{(8)}_{43j} = \sqrt{(c_{43}-c_j)^\top G_5 (c_{43}-c_j) + (r_{43}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{43}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{43} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds^2_{43} = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r^2_{43}\, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{43} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either

a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(43)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{43}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An expected value — a linear combination of two outcomes.

Structure: Exactly linear in the success probability, so speculation pays when $p > c/(g+c)$. Branch predictors raise p, which is why their accuracy matters quadratically for deep pipelines.

Reduction: Speculative gain is a linear expectation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

Distributed Systems

44. CAP Theorem

Origin (LA edition).

consistency, availability, partition tolerance: pick two

Input (8S edition).

$$\neg\,\text{Admit}_{44}(8S) \iff \eta^{(44)}_{\text{ind}} < \varepsilon_{\text{ind}} \;\vee\; \Delta_{8S} \leq \varepsilon_{\text{gain}}, \qquad e_{44} \in \mathcal{R}(C_4)$$

Calculus image.

$$\delta\Delta_{8S}[e_{44}] = 0 \;\text{ for every admissible variation } \iff (I - P_{\mathcal{V}})e_{44} = 0 \iff \det \text{Gr}(C_4, e_{44}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T14**, so the calculus image is of class **D14** (vanishing first variation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the vanishing first variation of Lemma D14, instantiated at the point $c_{44} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D14 needs nothing beyond Θ-A(3), the identity $\nabla_e \|w\|^2 = 2(I - P_{\mathcal{V}})e$. In particular no regularity, no bandwidth limit, and no property of the kernel is used — appropriately, since the claim is that there is no differential content to regularise.

(Q4) Derivation. Compute the first variation of the score in the direction of the entry's own content. By Θ-A(3) it is $2(I - P_{\mathcal{V}})e$, which vanishes identically precisely when $e \in \mathcal{V}$. Equivalently, by Θ-A(4), the Gram determinant

vanishes and the parallelepiped spanned by C_4 and e has zero five-dimensional volume. There is therefore nothing to differentiate, and the calculus image is the statement that the variation is identically zero.

(Q5) Retraction through the tower. On the lattice the vanishing variation becomes $\eta_{\text{ind}}^{(44)} = 0$, the 8S rejection certificate. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. The entry is **not** admitted, and the calculus image says why in a third vocabulary: the first variation of the score in the direction of this entry's content vanishes identically, so the Gâteaux derivative carries no information and the Gram volume $\det \mathrm{Gr}(C_4, e_{44})$ is zero. Differentiating an obstruction returns zero, and that is the correct output.

Objects: An impossibility result about three properties.

Structure: A proof by contradiction using a partitioned execution — combinatorial rather than algebraic. It is stated as a trilemma but is really a statement about behaviour during partitions, since partitions cannot be forgone.

Reduction: The theorem is a combinatorial impossibility; no linear structure applies.

Status: No faithful linear form.

8S status: Rejection certificate (Lemma T14) — the obstruction is preserved, not removed.

Calculus status: Vanishing first variation (Lemma D14) — the obstruction survives its third translation intact.

45. PACELC Extension

Origin (LA edition).

if Partition: A or C; Else: L or C

Input (8S edition).

$$\neg\,\mathrm{Admit}_{45}(8S) \iff \eta_{\text{ind}}^{(45)} < \varepsilon_{\text{ind}} \;\vee\; \Delta_{8S} \le \varepsilon_{\text{gain}}, \qquad e_{45} \in \mathcal{R}(C_4)$$

Calculus image.

$$\delta\Delta_{8S}[e_{45}] = 0 \;\text{ for every admissible variation } \iff (I - P_{\mathcal{V}})e_{45} = 0 \iff \det \mathrm{Gr}(C_4, e_{45}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T14**, so the calculus image is of class **D14** (vanishing first variation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the vanishing first variation of Lemma D14, instantiated at the point $c_{45} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D14 needs nothing beyond Θ-A(3), the identity $\nabla_e \|w\|^2 = 2(I - P_{\mathcal{V}})e$. In particular no regularity, no bandwidth limit, and no property of the kernel is used — appropriately, since the claim is that there is no differential content to regularise.

(Q4) Derivation. Compute the first variation of the score in the direction of the entry's own content. By Θ-A(3) it is $2(I - P_{\mathcal{V}})e$, which vanishes identically precisely when $e \in \mathcal{V}$. Equivalently, by Θ-A(4), the Gram determinant vanishes and the parallelepiped spanned by C_4 and e has zero five-dimensional volume. There is therefore nothing to differentiate, and the calculus image is the statement that the variation is identically zero.

(Q5) Retraction through the tower. On the lattice the vanishing variation becomes $\eta_{\text{ind}}^{(45)} = 0$, the 8S rejection certificate. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. The entry is **not** admitted, and the calculus image says why in a third vocabulary: the first variation of the score in the direction of this entry's content vanishes identically, so the Gâteaux derivative carries no information and the Gram volume $\det \mathrm{Gr}(C_4, e_{45})$ is zero. Differentiating an obstruction returns zero, and that is the correct output.

Objects: A refinement adding the non-partitioned case.

Structure: Makes explicit that the latency–consistency trade-off exists even without partitions — a second binary choice conditioned on the first. It is a decision tree, not a continuous trade-off.

Reduction: The extension is a conditional binary classification.

Status: Structural analogue.

8S status: Rejection certificate (Lemma T14) — the obstruction is preserved, not removed.

Calculus status: Vanishing first variation (Lemma D14) — the obstruction survives its third translation intact.

46. FLP Impossibility

Origin (LA edition).

no deterministic consensus with one faulty process in an asynchronous system

Input (8S edition).

$$\neg\,\mathrm{Admit}_{46}(8S) \iff \eta_{\mathrm{ind}}^{(46)} < \varepsilon_{\mathrm{ind}} \ \vee\ \Delta_{8S} \le \varepsilon_{\mathrm{gain}}, \qquad e_{46} \in \mathcal{R}(C_4)$$

Calculus image.

$$\delta\Delta_{8S}[e_{46}] = 0 \ \text{ for every admissible variation } \iff (I - P_{\mathcal{V}})e_{46} = 0 \iff \det \mathrm{Gr}(C_4, e_{46}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T14**, so the calculus image is of class **D14** (vanishing first variation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the vanishing first variation of Lemma D14, instantiated at the point $c_{46} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D14 needs nothing beyond Θ-A(3), the identity $\nabla_e \|w\|^2 = 2(I - P_{\mathcal{V}})e$. In particular no regularity, no bandwidth limit, and no property of the kernel is used — appropriately, since the claim is that there is no differential content to regularise.

(Q4) Derivation. Compute the first variation of the score in the direction of the entry's own content. By Θ-A(3) it is $2(I - P_{\mathcal{V}})e$, which vanishes identically precisely when $e \in \mathcal{V}$. Equivalently, by Θ-A(4), the Gram determinant vanishes and the parallelepiped spanned by C_4 and e has zero five-dimensional volume. There is therefore nothing to differentiate, and the calculus image is the statement that the variation is identically zero.

(Q5) Retraction through the tower. On the lattice the vanishing variation becomes $\eta_{\mathrm{ind}}^{(46)} = 0$, the 8S rejection certificate. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. The entry is **not** admitted, and the calculus image says why in a third vocabulary: the first variation of the score in the direction of this entry's content vanishes identically, so the Gâteaux derivative carries no information and the Gram volume $\det \mathrm{Gr}(C_4, e_{46})$ is zero. Differentiating an obstruction returns zero, and that is the correct output.

Objects: A bivalence argument on the space of executions.

Structure: The proof constructs an infinite execution that never decides, by showing a bivalent configuration always has a bivalent successor — a topological connectivity argument on the configuration space. It is one of

the deepest results in the field and has no matrix formulation.

Reduction: The proof is a connectivity argument on configuration space.

Status: No faithful linear form.

8S status: Rejection certificate (Lemma T14) — the obstruction is preserved, not removed.

Calculus status: Vanishing first variation (Lemma D14) — the obstruction survives its third translation intact.

47. Byzantine Fault Tolerance Bound

Origin (LA edition).

$$n \geq 3f + 1$$

Input (8S edition).

$$y_{47}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{47})\Big] z(u), \qquad z_{47}(u) = \begin{bmatrix} c_{47} \\ r_{47} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{47})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{47}\, \psi_{47}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{47} = -\mathcal{L}_\star \psi_{47}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{47} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{47})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(47)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{47}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A counting condition on honest majorities.

Structure: Two disjoint quorums of size $n - f$ must overlap in more than f nodes, which gives the bound by a counting argument. It is exactly a pigeonhole condition, and it appears again in entry 48.

Reduction: The bound is a counting (pigeonhole) argument.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

48. Quorum Intersection

Origin (LA edition).

$$|Q_1| + |Q_2| > n$$

Input (8S edition).

$$y_{48}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{48})\Big] z(u), \qquad z_{48}(u) = \begin{bmatrix} c_{48} \\ r_{48} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{48})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{48}\, \psi_{48}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{48} = -\mathcal{L}_\star \psi_{48}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{48} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{48})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(48)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{48}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An overlap condition on subsets.

Structure: Exactly the inclusion–exclusion condition guaranteeing two quorums share a node, which is what makes consistency possible. Weighted quorums generalise it to a linear inequality on weights.

Reduction: Intersection is an inclusion–exclusion inequality.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

49. Paxos and Raft Round Complexity

Origin (LA edition).

2 round trips in the common case

Input (8S edition).

$$\neg\,\mathrm{Admit}_{49}(8S) \iff \eta_{\mathrm{ind}}^{(49)} < \varepsilon_{\mathrm{ind}} \ \vee\ \Delta_{8S} \le \varepsilon_{\mathrm{gain}}, \qquad e_{49} \in \mathcal{R}(C_4)$$

Calculus image.

$$\delta\Delta_{8S}[e_{49}] = 0 \ \text{ for every admissible variation } \iff (I - P_{\mathcal{V}})e_{49} = 0 \iff \det \mathrm{Gr}(C_4, e_{49}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T14**, so the calculus image is of class **D14** (vanishing first variation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the vanishing first variation of Lemma D14, instantiated at the point $c_{49} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D14 needs nothing beyond Θ-A(3), the identity $\nabla_e \|w\|^2 = 2(I - P_{\mathcal{V}})e$. In particular no regularity, no bandwidth limit, and no property of the kernel is used — appropriately, since the claim is that there is no differential content to regularise.

(Q4) Derivation. Compute the first variation of the score in the direction of the entry's own content. By Θ-A(3) it is $2(I-P_{\mathcal{V}})e$, which vanishes identically precisely when $e \in \mathcal{V}$. Equivalently, by Θ-A(4), the Gram determinant vanishes and the parallelepiped spanned by C_4 and e has zero five-dimensional volume. There is therefore nothing to differentiate, and the calculus image is the statement that the variation is identically zero.

(Q5) Retraction through the tower. On the lattice the vanishing variation becomes $\eta_{\mathrm{ind}}^{(49)} = 0$, the 8S rejection certificate. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. The entry is **not** admitted, and the calculus image says why in a third vocabulary: the first variation of the score in the direction of this entry's content vanishes identically, so the Gâteaux derivative carries no information and the Gram volume $\det \mathrm{Gr}(C_4, e_{49})$ is zero. Differentiating an obstruction returns zero, and that is the correct output.

Objects: A message round count.

Structure: The prepare and accept phases each cost one round trip, and stable leadership eliminates the first — an amortisation. The counting is combinatorial, and the correctness rests on quorum intersection.

Reduction: Round counting is combinatorial; correctness rests on entry 48.

Status: Structural analogue.

8S status: Rejection certificate (Lemma T14) — the obstruction is preserved, not removed.

Calculus status: Vanishing first variation (Lemma D14) — the obstruction survives its third translation intact.

50. Lamport Logical Clock

Origin (LA edition).

$$a \to b \Rightarrow C(a) < C(b)$$

Input (8S edition).

$$y_{50}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{50})\Big] z(u), \qquad z_{50}(u) = \begin{bmatrix} c_{50} \\ r_{50}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{50})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{50}\, \psi_{50}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{50} = -\mathcal{L}_\star \psi_{50}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{50} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{50})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(50)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{50}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A monotone map from events to integers.

Structure: Consistent with the happens-before partial order but not complete — it embeds a partial order into a total one, which necessarily loses information. Concurrent events get ordered arbitrarily.

Reduction: The clock embeds a partial order into a total one, losing concurrency.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

51. Vector Clock

Origin (LA edition).

$$V(a) < V(b) \iff a \to b$$

Input (8S edition).

$$y_{51}(\tau,u)=\Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{51})\Big]z(u),\qquad z_{51}(u)=\begin{bmatrix}c_{51}\\ r_{51}u\end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{51})(c)=\int_{\mathcal{C}}\gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{51}\,\psi_{51}(c')\,p(c')\,dc',\qquad \partial_\tau\psi_{51}=-\mathcal{L}_\star\psi_{51},\qquad \mathcal{L}_\star=-\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{51}\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du=\sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{51})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S}-\nu(\varepsilon_{\text{ind}}-\eta^{(51)}_{\text{ind}})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{51}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A vector of per-process counters, ordered componentwise.

Structure: Exactly captures the partial order because the componentwise order on $\mathbb{N}^n$ is the right structure — incomparable vectors correspond to concurrent events. The cost is $O(n)$ space per timestamp, which is the price of representing a partial order faithfully.

Reduction: Vector clocks embed the causal partial order into the componentwise order on $\mathbb{N}^n$.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

52. Consistent Hashing

Origin (LA edition).

keys move only $O(K/n)$ on membership change

Input (8S edition).

$$E_{52}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 52}, \qquad \big\|z_{52}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta_{52j}^{(8)} = \sqrt{(c_{52}-c_j)^\top G_5 (c_{52}-c_j) + (r_{52}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{52}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{52} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{52}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{52}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{52} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(52)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{52}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A mapping of keys onto a circular space.

Structure: Placing nodes and keys on a ring means adding a node only steals from its successor — a locality property of the circular order. Virtual nodes smooth the load by sampling the ring more finely, reducing the variance of the partition sizes.

Reduction: Consistent hashing exploits locality of a circular ordering.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

53. CRDT Convergence

Origin (LA edition).

commutative, associative, idempotent merge

Input (8S edition).

$$\mathcal{Y}_{53}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{53})\Big] z(u), \qquad z_{53}(u) = \begin{bmatrix} c_{53} \\ r_{53} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{53})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{53}\, \psi_{53}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{53} = -\mathcal{L}_\star \psi_{53}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{53} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{53})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(53)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{53}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A join-semilattice.

Structure: Convergence follows because the merge operation is a least upper bound in a lattice, so applying updates in any order reaches the same supremum. It is an algebraic guarantee — replicas converge without coordination because the algebra forces it.

Reduction: CRDTs converge because merges are lattice joins.

Status: Exact linear (lattice).

8S status: Exact linear (lattice); preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear (lattice); carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

54. Two-Phase Commit Blocking

Origin (LA edition).

coordinator failure blocks participants

Input (8S edition).

$$\neg\,\text{Admit}_{54}(8S) \iff \eta^{(54)}_{\text{ind}} < \varepsilon_{\text{ind}} \ \vee\ \Delta_{8S} \leq \varepsilon_{\text{gain}}, \qquad e_{54} \in \mathcal{R}(C_4)$$

Calculus image.

$$\delta\Delta_{8S}[e_{54}] = 0 \ \text{ for every admissible variation } \iff (I - P_{\mathcal{V}})e_{54} = 0 \iff \det \text{Gr}(C_4, e_{54}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T14**, so the calculus image is of class **D14** (vanishing first variation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the vanishing first variation of Lemma D14, instantiated at the point $c_{54} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D14 needs nothing beyond Θ-A(3), the identity $\nabla_e \|w\|^2 = 2(I - P_{\mathcal{V}})e$. In particular no regularity, no bandwidth limit, and no property of the kernel is used — appropriately, since the claim is that there is no differential content to regularise.

(Q4) Derivation. Compute the first variation of the score in the direction of the entry's own content. By Θ-A(3) it is $2(I-P_{\mathcal{V}})e$, which vanishes identically precisely when $e \in \mathcal{V}$. Equivalently, by Θ-A(4), the Gram determinant vanishes and the parallelepiped spanned by C_4 and e has zero five-dimensional volume. There is therefore nothing to differentiate, and the calculus image is the statement that the variation is identically zero.

(Q5) Retraction through the tower. On the lattice the vanishing variation becomes $\eta^{(54)}_{\text{ind}} = 0$, the 8S rejection certificate. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. The entry is **not** admitted, and the calculus image says why in a third vocabulary: the first variation of the score in the direction of this entry's content vanishes identically, so the Gâteaux derivative carries no information and the Gram volume $\det \mathrm{Gr}(C_4, e_{54})$ is zero. Differentiating an obstruction returns zero, and that is the correct output.

Objects: A protocol with a state in which progress is impossible.

Structure: Participants that have voted but not learned the outcome cannot safely proceed — an uncertainty state in the protocol's state machine. Three-phase commit adds a state to break it, at the cost of extra rounds and weaker fault assumptions.

Reduction: Blocking is an uncertainty state in the protocol automaton.

Status: Structural analogue.

8S status: Rejection certificate (Lemma T14) — the obstruction is preserved, not removed.

Calculus status: Vanishing first variation (Lemma D14) — the obstruction survives its third translation intact.

Scheduling, Reliability, and Energy

55. Load Imbalance Factor

Origin (LA edition).

$$\lambda = \frac{\max_i T_i}{\bar{T}}$$

Input (8S edition).

$$\left(g^{(5)}_{55j},\ \delta^{(8)}_{55j},\ g^{(3)}_{55j},\ g^{(\mathrm{J})}_{55j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{55j}, \qquad g^{(3)}_{55j} \le g^{(5)}_{55j}$$

Calculus image.

$$d_g(z_{55}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{55j} \le d_g(z_{55}, z_j), \qquad g^{(3)}_{55j} \le g^{(5)}_{55j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no

part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{55} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{55j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(55)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{55}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio of maximum to mean — an ℓ_∞ over ℓ_1 comparison.

Structure: Parallel time is governed by the maximum, not the mean, so imbalance directly multiplies the runtime. It is exactly the ratio of two norms of the work vector, and reducing it is a load balancing (partitioning) problem.

Reduction: Imbalance is an ℓ_∞/ℓ_1 norm ratio of the work vector.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

56. Work Stealing Bound

Origin (LA edition).

$$T_p \le \frac{W}{p} + O(D)$$

Input (8S edition).

$$y_{56}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{56})\Big] z(u), \qquad z_{56}(u) = \begin{bmatrix} c_{56} \\ r_{56} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{56})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{56}\, \psi_{56}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{56} = -\mathcal{L}_\star \psi_{56}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{56} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{56})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(56)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{56}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Brent's bound achieved by a randomised scheduler.

Structure: Randomised stealing achieves the work–depth bound in expectation, with the steal count bounded by $O(pD)$ — a probabilistic argument on the number of failed steals. The bound is tight up to constants.

Reduction: Work stealing achieves the affine work–depth bound in expectation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

57. List Scheduling Bound

Origin (LA edition).

$$T \leq (2 - \frac{1}{p}) T_{\text{opt}}$$

Input (8S edition).

$$\text{Admit}_{57}(8S) \iff \eta_{\text{ind}}^{(57)} \geq \varepsilon_{\text{ind}} \;\wedge\; \Delta_{8S} > \varepsilon_{\text{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \;\text{ or }\; \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{57}(x, \lambda) = 0, \qquad \partial_x \mathcal{G}_{57} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x \mathcal{G}_{57} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{57} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x, \lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(57)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{57}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A worst-case approximation ratio.

Structure: Graham's bound follows from two lower bounds — total work over p and the critical path — and the schedule cannot exceed their sum. It is exactly a sum of two linear lower bounds, and the factor two comes from combining them.

Reduction: The bound sums two linear lower bounds.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear; carried to the continuum by Lemma D7, and the input was algebraic, so the differential content is entirely supplied by the transform.

58. Checkpoint Interval (Young–Daly)

Origin (LA edition).

$$T_{\text{opt}} = \sqrt{2\,C\,\text{MTBF}}$$

Input (8S edition).

$$\max_{n_{58} \in \mathbb{Z}^5} \; \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{58}\|_1 \leq M, \;\; \eta^{(58)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{58}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(58)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_{58}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(58)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{58} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(58)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{58}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A minimiser of a sum of two opposing costs.

Structure: Checkpoint overhead falls as the interval grows while expected lost work rises, so minimising $C/T + T/(2\,\text{MTBF})$ gives the square root — the same structure as the economic order quantity. It is a one-variable convex minimisation.

Reduction: The interval minimises a sum of a reciprocal and a linear term.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

59. System Mean Time Between Failures

Origin (LA edition).

$$\text{MTBF}_{\text{sys}} = \frac{\text{MTBF}_{\text{node}}}{n}$$

Input (8S edition).

$$y_{59}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{59})\Big] z(u), \qquad z_{59}(u) = \begin{bmatrix} c_{59} \\ r_{59} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{59})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{59}\, \psi_{59}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{59} = -\mathcal{L}_\star \psi_{59}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{59} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind.

Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{59})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(59)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{59}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Failure rates adding across components.

Structure: Rates add for series systems, so the reciprocal times combine harmonically — exactly the same rule as parallel resistances and effective population size. At exascale this makes failures a routine event rather than an exception.

Reduction: Failure rates add; MTBFs combine harmonically.

Status: Exact linear in reciprocals.

8S status: Exact linear in reciprocals; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear in reciprocals; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

60. Energy–Delay Product and Power Scaling

Origin (LA edition).

$$P \propto CV^2 f, \qquad \text{EDP} = E \times T$$

Input (8S edition).

$$\log r_{60} = \log r_{\max} + \alpha \log B_5(c_{60}) + \tfrac{1}{3} \log s_{60}, \qquad \log B_5(c_{60}) = \sum_{a=4}^{8} \log b(c_{60,a})$$

Calculus image.

$$d \log r_{60} = \alpha\, d \log B_5(c_{60}) + \tfrac{1}{3}\, d \log s_{60}, \qquad d(d \log r_{60}) = 0, \qquad \left.\frac{\partial \log r_{60}}{\partial \log b(c_{60,a})}\right|_{s_{60}} = \alpha \quad (a = 4, \dots, 8)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T6**, so the calculus image is of class **D6** (exact differential form and elasticity). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the exact differential form and elasticity of Lemma D6, instantiated at the point $c_{60} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D6 needs $B_5(c) > 0$ and $s > 0$ so the logarithms and the form are defined; by Theorem I(4) of the 8S edition this fails exactly at a bump zero or under full suppression, where $r = 0$ and the fibre degenerates. Exactness needs only that $\log r$ be a well-defined C^1 function, which it is on that region.

(Q4) Derivation. Take logarithms of $r = r_{\max} B_5^{\alpha} s^{1/3}$ and of $B_5 = \prod_{a=4}^{8} b(c_a)$, then differentiate. Since the result is the differential of a function, the form is exact; exact forms are closed by $d \circ d = 0$, and their line integrals depend only on the endpoints by the fundamental theorem of calculus for line integrals. The elasticities are read off as the coefficients — as partial derivatives in the factorised coordinates $(\log b(c_4), \dots, \log b(c_8), \log s)$, with the

other factors held fixed. If the salience s itself varies over the base, the *total* derivative along c_a carries the extra term $\frac{1}{3}\partial_a \log s$ and is not α; Θ-I(2) is stated in the factorised coordinates precisely to keep that distinction visible.

(Q5) Retraction through the tower. Integrating the form between two lattice points returns the 8S difference of logarithms, $\log r_{60} - \log r_j$, exactly and with no quadrature error, because the form is exact. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission is the first-order condition $\delta\Delta_{8S} = 0$ subject to $\eta_{\text{ind}}^{(60)} \geq \varepsilon_{\text{ind}}$, together with the domain condition $B_5(c_{60}) > 0$ and $s_{60} > 0$ that makes the logarithms and the form defined. Where the domain condition fails, $r_{60} = 0$, the fibre collapses to a point, and ds^2 degenerates to the base metric.

Objects: A multiplicative power model and a product metric.

Structure: Since $f \propto V$ roughly, power scales as the cube of frequency and energy per operation as the square of voltage — so slowing down saves energy superlinearly. The energy–delay product is a product of two competing quantities, and minimising it is a log-space optimisation where the two log-linear trends cross.

Reduction: Power is multiplicative; the EDP optimum is where two log-linear trends balance.

Status: Exact linear in logs.

8S status: Exact linear in logs; preserved under the transform by Theorem C and Lemma T6.

Calculus status: Exact linear in logs; carried to the continuum by Lemma D6, and the input was algebraic, so the differential content is entirely supplied by the transform.